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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a comprehensive and detailed description of the Library Management System (LMS), version 1.0. This document aims to clearly define the functional and non-functional requirements of the system, ensuring a shared understanding among all stakeholders involved in the development process.

The scope of this SRS includes the complete functionality of the LMS, covering modules such as book inventory management, user account handling, borrowing and return processes, and reporting mechanisms. The document serves as a formal agreement between stakeholders, including clients, developers, and project managers, regarding what the system is expected to accomplish.

Additionally, this SRS will act as a reference point throughout the software development lifecycle, supporting activities such as system design, implementation, testing, and maintenance. It is intended to minimize ambiguity, reduce development risks, and ensure that the final system aligns with user expectations and organizational objectives.

1.2 Document Conventions

This document follows standard IEEE documentation conventions to ensure clarity, consistency, and readability. The structure is organized into hierarchical sections and subsections, each identified by numbered headings.

The following conventions are used throughout the document:

- **Bold text** is used for section headings and important terms to improve readability.
- The term “shall” is used to indicate mandatory system requirements that must be fulfilled.
- The term “should” is used to indicate recommended requirements that are desirable but not strictly mandatory.
- The term “may” is used to indicate optional features or functionalities.

Each functional requirement is uniquely identified using a standardized labeling format (e.g., REQ-LMS-01, REQ-LMS-02) to ensure traceability and ease of reference during development and testing phases.

Priority levels are assigned to requirements as follows:

- High (H): Critical for system operation
- Medium (M): Important but not critical
- Low (L): Optional or enhancement features

1.3 Intended Audience and Reading Suggestions

This document is intended for a diverse group of stakeholders involved in the development, deployment, and use of the Library Management System. Each group may refer to different sections of the document depending on their role and requirements.

Intended Audience:

- **Developers:** Responsible for implementing the system based on defined requirements
- **Project Managers:** Oversee planning, scheduling, and progress tracking
- **System Testers:** Validate the system against specified requirements
- **End Users (Librarians, Students, Faculty):** Interact with the system in daily operations
- **Documentation Writers:** Prepare user manuals and help guides

Reading Suggestions:

To effectively understand the system, readers are advised to follow a structured approach:

- Begin with **Section 1 (Introduction)** to understand the purpose and scope
- Proceed to **Section 2 (Requirements Gathering & Analysis)** for insights into requirement origins
- Review **Section 3 (Overall Description)** for a high-level system overview
- Focus on **Section 5 (System Features)** for detailed functional requirements
- Refer to **Appendices** for diagrams and supporting materials

Different readers may prioritize sections based on their roles. For example, developers should carefully study the entire document, while end users may focus more on system features and user interface descriptions.

1.4 Product Scope

The Library Management System (LMS) is a software solution designed to automate and optimize the core operations of a library. The system aims to replace traditional manual processes with an efficient, reliable, and user-friendly digital platform.

The primary objective of the LMS is to streamline activities such as book cataloging, user registration, borrowing and returning of books, and fine calculation. By centralizing all library-related data into a unified system, the LMS enhances accuracy, reduces redundancy, and improves overall operational efficiency.

Key Goals:

- To provide a centralized database for managing books and users
- To enable quick and efficient searching of library resources

- To automate transaction processes such as issuing and returning books
- To minimize human errors associated with manual record keeping

Expected Benefits:

- Improved efficiency and time savings
- Enhanced user experience through intuitive interfaces
- Better tracking and management of library assets
- Increased reliability and data consistency

The LMS aligns with organizational goals of digital transformation and improved service delivery, making it a valuable asset for modern educational institutions.

1.5 References

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- [2] Goldman, J. and Song, I. (2005). "Organizing and Managing Use Cases," in the *First International Workshop on Best Practices of UML*, Oct. 2628, 2005, Klagenfurt, Austria, (in *Perspectives in Conceptual Modeling, LNCS Vol. 3770*, Editor: J. Akoka, etc. Springer 11 Verlag, 2005), pp. 4352
- [3] D. A. A. Hussain, "Design and Implementation of Library Management System," B.Sc. thesis, Dept. of Computer Engineering, College of Engineering, Al-Nahrain University, Baghdad, Iraq, 2025.
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2. Requirements Gathering & Analysis

2.1 Customer Interaction Overview

The requirements for the Library Management System (LMS) were elicited through structured and simulated interactions with key stakeholders representing the primary user groups of the system. Since this is an academic exercise, these interactions were modeled to reflect real-world scenarios typically encountered in library environments.

The stakeholders involved in the requirement gathering process include:

- Librarians, who are responsible for managing daily library operations
- Students, who are the primary users of library services
- Faculty members, who utilize the library for academic and research purposes
- System administrators, who oversee system maintenance and control

Each stakeholder group contributes unique requirements based on their roles and expectations. For instance, librarians emphasized the need for efficient book tracking and inventory management, while students highlighted the importance of quick search and easy borrowing processes.

The interaction process ensured that both functional and non-functional requirements were captured comprehensively. It also helped in identifying potential challenges and usability concerns that could arise during system implementation.

2.2 Interview Method

A structured interview methodology was adopted to gather detailed and consistent information from stakeholders. A predefined set of questions was prepared to guide the interviews and ensure that all relevant aspects of the system were addressed.

Objectives of the Interview Process:

- To understand the limitations of existing systems
- To identify user expectations and requirements
- To gather insights into daily operational challenges
- To define system functionalities from a user-centric perspective

Sample Interview Questions:

- What difficulties do you face while managing or using the current library system?
- How do you currently track issued and returned books?
- What features would you like to see in a new system?
- How should overdue books and fines be handled?

Summary of Responses:

- Librarians expressed the need for automation to reduce manual workload and errors
- Students emphasized the importance of fast and accurate search functionality
- Faculty members required extended borrowing privileges and better tracking
- Administrators highlighted the need for reporting and monitoring capabilities

The responses were carefully analyzed and categorized into functional requirements (system behavior) and non-functional requirements (performance, usability, security, etc.).

2.3 Brainstorming Session

In addition to interviews, brainstorming sessions were conducted among the development team to generate ideas and refine system requirements. This collaborative approach encouraged creative thinking and helped identify additional features that could enhance the system.

During the brainstorming sessions:

- Team members proposed various features and improvements
- Ideas were evaluated based on feasibility, cost, and impact
- Redundant or impractical suggestions were filtered out

Key Outcomes:

- Inclusion of advanced search filters (by category, author, availability)
- Implementation of automated fine calculation
- Development of a user-friendly interface
- Addition of reporting and analytics features

The brainstorming process played a crucial role in expanding the scope of the system while ensuring that it remains practical and implementable.

2.4 Observation Study

An observation study was conducted to analyze the workflow of existing library systems. Both manual systems (register-based) and semi-automated systems (spreadsheet-based) were examined.

Observed Processes:

- Manual entry of book records in registers
- Recording of issue and return transactions
- Calculation of fines using basic methods

Key Findings:

- Manual systems are time-consuming and prone to human errors
- Data retrieval is inefficient and often delayed
- Lack of centralized data storage leads to inconsistencies
- Tracking overdue books is difficult and unreliable

These observations highlighted the need for a centralized, automated system that can handle large volumes of data efficiently and accurately.

2.5 Business Problem Description

The existing library management approach suffers from several critical issues that negatively impact efficiency and user satisfaction.

Identified Problems:

- Dependence on manual processes for record keeping
- Difficulty in maintaining accurate and up-to-date records
- Inefficient search and retrieval of book information
- Lack of real-time tracking of issued and returned books
- Errors in fine calculation and reporting

These problems lead to increased workload for library staff, delays in service delivery, and dissatisfaction among users. The absence of a centralized system further exacerbates these issues by causing data inconsistencies and duplication.

The proposed Library Management System aims to address these challenges by providing an integrated, automated solution that improves efficiency, accuracy, and user experience.

2.6 Expected System Features

Based on the requirement analysis, the LMS is expected to include the following key features:

- A comprehensive book management system for adding, updating, and deleting records
- An efficient search mechanism to locate books quickly using multiple criteria
- A user management module to handle registration and authentication
- An automated issue and return system to track transactions
- A fine calculation mechanism for overdue books
- Reporting and analytics features for monitoring system usage

These features are designed to meet the needs of all stakeholders and ensure smooth library operations.

2.7 Feasibility & Requirement Analysis

Technical Feasibility

The system can be developed using widely available technologies such as web-based frameworks, relational databases, and standard programming languages. The required tools and expertise are readily accessible, making the project technically feasible.

Economic Feasibility

The cost of developing and maintaining the system is relatively low compared to the benefits it provides. Automation reduces manual labor, minimizes errors, and improves efficiency, resulting in long-term cost savings.

Operational Feasibility

The LMS is designed to be user-friendly and intuitive, ensuring that users can easily adapt to the system with minimal training. The system integrates well with existing workflows, making it operationally feasible.

3. Overall Description

3.1 Product Perspective

The Library Management System is designed as a standalone application that replaces traditional manual library systems. It serves as a centralized platform for managing all library-related activities, including book inventory, user records, and transactions.

The system can also be extended in the future to integrate with larger institutional systems such as:

- Student Information Systems (SIS)
- Enterprise Resource Planning (ERP) systems
- Online digital libraries

The LMS interacts with external entities such as users (students, faculty) and administrators, forming a complete ecosystem for library management.

3.2 Product Functions

The LMS provides a wide range of functions that collectively support efficient library operations. These functions are summarized below:

- **Book Management:** Add, update, delete, and categorize books
- **User Management:** Register users and manage user profiles
- **Transaction Management:** Issue and return books, track due dates
- **Fine Management:** Calculate and record fines for overdue books
- **Search Functionality:** Locate books using various search criteria
- **Reporting:** Generate reports on system usage and book availability

These functions are interconnected and work together to provide a seamless user experience.

3.3 User Classes and Characteristics

The system supports multiple user classes, each with distinct roles, responsibilities, and levels of access.

Administrator

The administrator has full control over the system and is responsible for managing all aspects of the LMS.

Characteristics:

- High level of technical expertise
- Authorized to configure system settings
- Access to all data and reports

Librarian

The librarian is the primary operator of the system and handles day-to-day library operations.

Characteristics:

- Moderate technical knowledge
- Responsible for managing books and transactions
- Requires efficient and easy-to-use interfaces

Student/User

Students are the primary users of the system who interact with it to search and borrow books.

Characteristics:

- Basic computer knowledge
- Frequent usage of search and borrowing features
- Limited access rights

Faculty Member

Faculty members use the system for academic and research purposes.

Characteristics:

- Similar to students but may have extended privileges
- Requires access to a larger number of resources

3.4 Operating Environment

The LMS is designed to operate in a web-based environment, ensuring accessibility across multiple platforms.

Environment Details:

- Operating Systems: Windows, Linux, macOS
- Web Browsers: Google Chrome, Microsoft Edge, Mozilla Firefox
- Hardware: Standard computer systems with internet connectivity

The system is designed to function efficiently under normal network conditions and support multiple users simultaneously.

3.5 Design and Implementation Constraints

The development of the LMS is subject to several constraints that may influence design and implementation decisions.

Constraints Include:

- Use of a relational database system (e.g., MySQL)
- Compliance with security and data protection standards
- Limited development time and budget
- Requirement to use standard web technologies

These constraints must be carefully considered to ensure successful project completion.

3.6 User Documentation

To ensure effective usage of the system, comprehensive user documentation will be provided.

Documentation Includes:

- User manuals explaining system features
- Step-by-step guides for common tasks
- Online help system integrated into the application
- Tutorials for new users

The documentation will be designed to be clear, concise, and accessible to users with varying levels of technical expertise.

3.7 Assumptions and Dependencies

Assumptions:

- Users have basic knowledge of computer operations
- Reliable internet connectivity is available
- Users have valid login credentials

Dependencies:

- The system depends on the availability of a database server
- Integration with other systems may be required in the future
- External libraries or frameworks may be used during development

Any changes to these assumptions or dependencies may impact the system's functionality and performance.

4. External Interface Requirements

4.1 User Interfaces

The Library Management System (LMS) provides multiple user interfaces designed to facilitate interaction between the system and its users, including administrators, librarians, students, and faculty members. The user interface is designed with a focus on usability, accessibility, and efficiency, ensuring that users with varying levels of technical expertise can effectively operate the system.

General UI Characteristics

The system will adopt a web-based graphical user interface (GUI) accessible through standard web browsers. The interface will follow consistent design principles to maintain uniformity across all screens.

Key characteristics include:

- Clean and intuitive layout with minimal complexity
- Consistent color scheme and typography
- Responsive design to support different screen sizes
- Clear navigation structure with menus and submenus

Primary User Interface Components

1. Login Interface

- Allows users to enter credentials (username and password)
- Includes options such as “Forgot Password” and “Reset Password”
- Displays appropriate error messages for invalid login attempts

2. Dashboard Interface

- Provides an overview of system activities
- Displays key information such as number of available books, issued books, and pending returns
- Includes quick access links to major functionalities

3. Book Management Interface

- Enables librarians to add, update, and delete book records
- Displays book details in tabular format
- Supports sorting and filtering of records

4. Search Interface

- Allows users to search for books using multiple criteria such as title, author, ISBN, or category
- Displays search results dynamically
- Provides filters to refine search results

5. Issue/Return Interface

- Allows librarians to issue and return books
- Displays due dates and calculates fines automatically
- Provides confirmation messages for successful transactions

Standard UI Features

- Navigation bar present on all screens
- Standard buttons such as “Submit,” “Cancel,” “Edit,” and “Delete”
- Help and support option available on all pages
- Confirmation dialogs for critical actions

Error Handling and Messages

- Clear and descriptive error messages
- Validation of input fields (e.g., required fields, correct formats)
- Alerts for invalid operations

Accessibility and Usability Considerations

- Simple language for ease of understanding
- Keyboard navigation support
- Minimal number of steps for completing tasks

Detailed UI design specifications, including mockups and layouts, will be documented separately in the User Interface Design Document.

4.2 Hardware Interfaces

The LMS interacts with various hardware components to support its functionality. These interfaces define how the software communicates with physical devices.

Supported Hardware Components

- Desktop computers and laptops
- Servers for hosting the application and database
- Barcode scanners (optional) for book identification
- Printers for generating reports and receipts

Nature of Interaction

- Input devices (keyboard, mouse) are used for user interaction
- Output devices (monitor, printer) display information and reports
- Barcode scanners can be used to quickly input book identification data

Data and Control Interaction

- The system receives input data from users through input devices
- Processes data internally and sends output to display or print devices
- Communication between hardware and software is managed through standard device drivers

Hardware Constraints

- Requires systems with minimum processing capability and memory
- Stable power supply and network connectivity required

4.3 Software Interfaces

The LMS interacts with several software components to ensure proper functionality and data management.

Database Interface

- The system uses a relational database management system (RDBMS), such as MySQL
- Stores data related to books, users, and transactions
- Supports structured query language (SQL) for data operations

Operating System Interface

- Compatible with operating systems such as Windows and Linux
- Relies on OS-level services for file handling and process management

Web Browser Interface

- Users access the system through web browsers such as Google Chrome, Mozilla Firefox, or Microsoft Edge
- The system uses standard web technologies (HTML, CSS, JavaScript)

External Software Components

- May integrate with authentication services (if extended)
- May use third-party libraries for UI design or reporting

Data Exchange

- Input data: User credentials, book details, transaction records
- Output data: Reports, search results, notifications

Data is exchanged in structured formats and stored securely within the system.

Communication Between Components

- Frontend communicates with backend using HTTP/HTTPS
- Backend interacts with database using SQL queries
- Data is processed and returned in structured formats (e.g., JSON/XML if applicable)

4.4 Communications Interfaces

The LMS requires communication interfaces to enable interaction over networks and support data exchange between system components.

Network Communication

- The system operates over Local Area Network (LAN) or the Internet
- Uses standard communication protocols such as HTTP and HTTPS

Web-Based Communication

- All user interactions occur through a web browser
- Requests are sent to the server and responses are returned dynamically

Data Transfer

- Data is transmitted between client and server in secure formats
- Efficient data transfer mechanisms are used to ensure quick response times

Security and Encryption

- HTTPS protocol is used to ensure secure communication
- User data is encrypted during transmission
- Authentication mechanisms prevent unauthorized access

Email and Notifications (Optional Feature)

- The system may support email notifications for due dates and alerts
- Standard email protocols (SMTP) may be used

Performance Considerations

- The system should support multiple concurrent users

- Network latency should be minimized
- Data synchronization must be maintained across sessions

5. System Features

This section describes the major functional features of the Library Management System (LMS). Each feature represents a key service provided by the system and is described in terms of its purpose, behavior, and detailed functional requirements.

5.1 System Feature 1

5.1.1 Description and Priority

The Book Management feature enables librarians and administrators to manage the library's book inventory. This includes adding new books, updating existing records, and removing obsolete entries.

Priority: High

Benefit: 9

Penalty: 8

Cost: 5

Risk: 4

This feature is critical as it forms the foundation of the system.

5.1.2 Stimulus/Response Sequences

Stimulus	System Response
Librarian enters new book details	System validates and stores the data
Librarian updates book record	System updates database
Librarian deletes book	System removes record after confirmation
Invalid input entered	System displays error message

5.1.3 Functional Requirements

- **REQ-LMS-BM-01:** The system shall allow the librarian to add new books with details such as title, author, ISBN, category, and quantity.
- **REQ-LMS-BM-02:** The system shall validate all input fields before storing data.
- **REQ-LMS-BM-03:** The system shall allow modification of existing book records.
- **REQ-LMS-BM-04:** The system shall allow deletion of book records with confirmation.
- **REQ-LMS-BM-05:** The system shall maintain the availability status of each book.
- **REQ-LMS-BM-06:** The system shall display error messages for invalid or incomplete input.

5.2 User Management

5.2.1 Description and Priority

This feature manages user accounts including registration, updating user details, and role assignment.

Priority: High

Benefit: 9

Penalty: 7

Cost: 4

Risk: 3

5.2.2 Stimulus/Response Sequences

Stimulus	Response
User registers	System creates account
Admin updates user info	System updates database
Invalid login	Error message displayed

5.2.3 Functional Requirements

- **REQ-LMS-UM-01:** The system shall allow new users to register with valid credentials.
- **REQ-LMS-UM-02:** The system shall allow administrators to update user details.
- **REQ-LMS-UM-03:** The system shall assign roles (admin, librarian, user).
- **REQ-LMS-UM-04:** The system shall authenticate users during login.
- **REQ-LMS-UM-05:** The system shall prevent duplicate user registrations.

5.3 Book Issue and Return

5.3.1 Description and Priority

This feature manages borrowing and returning of books.

Priority: High

Benefit: 10

Penalty: 9

Cost: 6

Risk: 5

5.3.2 Stimulus/Response Sequences

Stimulus	Response
User requests book	System checks availability
Book issued	System updates record
Book returned	System updates status
Late return	System calculates fine

5.3.3 Functional Requirements

- **REQ-LMS-IR-01:** The system shall allow issuing of available books.
- **REQ-LMS-IR-02:** The system shall record issue date and due date.
- **REQ-LMS-IR-03:** The system shall allow book return processing.
- **REQ-LMS-IR-04:** The system shall calculate fines for overdue books.
- **REQ-LMS-IR-05:** The system shall restrict issuing if limit exceeded.

5.4 Search Functionality

5.4.1 Description and Priority

This feature enables users to search for books efficiently.

Priority: High

Benefit: 9

Penalty: 8

Cost: 4

Risk: 3

5.4.2 Stimulus/Response Sequences

Stimulus	Response
User enters search query	System displays results
No results found	System shows message

5.4.3 Functional Requirements

- **REQ-LMS-SR-01:** The system shall allow search by title, author, and ISBN.
- **REQ-LMS-SR-02:** The system shall display search results instantly.
- **REQ-LMS-SR-03:** The system shall allow filtering of results.
- **REQ-LMS-SR-04:** The system shall display message if no results found.

5.2 Fine Management

5.5.1 Description and Priority

Handles fine calculation for overdue books.

Priority: Medium

Benefit: 7

Penalty: 6

Cost: 3

Risk: 2

5.5.2 Stimulus/Response Sequences

Stimulus	Response
Book overdue	System calculates fine
Fine paid	System updates record

5.5.3 Functional Requirements

- **REQ-LMS-FN-01:** The system shall calculate fine based on delay.
- **REQ-LMS-FN-02:** The system shall update fine records.
- **REQ-LMS-FN-03:** The system shall display fine details to user.

5.6 Report Generation

5.6.1 Description and Priority

Generates reports for analysis.

Priority: Medium

5.6.2 Functional Requirements

- **REQ-LMS-RP-01:** Generate reports on issued books
- **REQ-LMS-RP-02:** Generate user activity reports

6. Other Nonfunctional Requirements

Non-functional requirements define the quality attributes and constraints under which the Library Management System (LMS) must operate. These requirements do not describe specific system behaviors but instead specify criteria that judge the system's performance, reliability, security, and usability.

6.1 Performance Requirements

The performance requirements specify how efficiently the system should operate under various conditions. These requirements ensure that the LMS provides a smooth and responsive user experience even during peak usage.

Response Time

- The system shall respond to user actions (such as search queries or login requests) within **2 seconds** under normal operating conditions.
- For complex operations such as report generation, the response time shall not exceed **5 seconds**.

Concurrency

- The system shall support a minimum of **100 concurrent users** without significant degradation in performance.
- The system shall maintain consistent performance levels during peak usage hours.

Throughput

- The system shall be capable of processing multiple transactions (such as issuing or returning books) simultaneously.
- Database operations shall be optimized to ensure minimal delay in data retrieval and updates.

Scalability

- The system shall be scalable to accommodate an increasing number of users and records without requiring major architectural changes.

Rationale

These requirements ensure that the system remains efficient and responsive, thereby improving user satisfaction and operational productivity.

6.2 Safety Requirements

Safety requirements focus on preventing data loss, system failures, and unintended actions that could harm the system or its users.

Data Protection

- The system shall implement regular data backup mechanisms to prevent loss of critical information.
- Backup operations shall be performed automatically at scheduled intervals.

Error Handling

- The system shall detect and handle errors gracefully without crashing.
- In case of system failure, appropriate error messages shall be displayed to the user.

Data Integrity

- The system shall ensure that all transactions are completed accurately and consistently.
- Partial or incomplete transactions shall be rolled back to maintain system consistency.

Preventive Measures

- The system shall prevent accidental deletion of important records by requiring user confirmation.
- Critical operations shall be restricted to authorized users only.

Rationale

These safety measures ensure the reliability of the system and protect against potential data loss or corruption.

6.3 Security Requirements

Security requirements define how the system protects user data and prevents unauthorized access.

Authentication

- The system shall require users to log in using valid credentials (username and password).

- Passwords shall be stored in encrypted form.

Authorization

- The system shall implement role-based access control (RBAC).
- Different user roles (admin, librarian, student) shall have different access privileges.

Data Security

- Sensitive data shall be encrypted during transmission using secure protocols (e.g., HTTPS).
- The system shall prevent unauthorized access to confidential information.

Session Management

- The system shall automatically log out users after a period of inactivity.
- The system shall prevent session hijacking through secure session handling.

Audit and Monitoring

- The system shall maintain logs of user activities for monitoring and auditing purposes.
- Unauthorized access attempts shall be recorded and flagged.

Rationale

Security requirements are critical to protect sensitive user data and maintain trust in the system.

6.4 Software Quality Attributes

This section defines the quality characteristics that the LMS must exhibit to ensure long-term usability and maintainability.

Reliability

- The system shall operate continuously without failure for extended periods.
- The system shall recover quickly from errors or failures.

Usability

- The system shall provide an intuitive and user-friendly interface.
- Users shall be able to perform common tasks with minimal training.

Maintainability

- The system shall be designed using modular architecture to facilitate easy updates and modifications.
- Code shall be well-documented and structured.

Scalability

- The system shall support future enhancements such as additional features or increased user load.

Portability

- The system shall be compatible with multiple operating systems and web browsers.

Testability

- The system shall be designed in a way that allows easy testing of individual components.

Availability

- The system shall be available for use at least 99% of the time, excluding scheduled maintenance.

Rationale

These attributes ensure that the system remains efficient, adaptable, and easy to maintain over time.

6.5 Business Rules

Business rules define the operational policies and constraints that govern the functioning of the LMS.

Borrowing Rules

- A user shall be allowed to borrow a maximum of **3** books at a time.
- The borrowing period shall be limited to 14 days.

Fine Policy

- A fine of ₹5 per day shall be charged for overdue books.
- Fines must be cleared before issuing new books.

User Roles

- Only librarians shall have permission to issue and return books.
- Only administrators shall have permission to modify system configurations.

Book Management Rules

- Books must be uniquely identified using ISBN or system-generated IDs.
- Duplicate entries shall not be allowed.

Operational Rules

- Users must have valid accounts to access system features.
- Inactive accounts may be deactivated after a specified period.

Rationale

These rules ensure consistent system behavior and align the software with organizational policies and procedures.

7. Other Requirements

This section specifies additional requirements that are not covered in earlier sections of the Software Requirements Specification. These requirements address supporting aspects such as database design, legal considerations, scalability, and future enhancements.

7.1 Database Requirements

The Library Management System shall use a structured relational database to store and manage all system data efficiently.

Database Characteristics

- The system shall use a Relational Database Management System (RDBMS) such as MySQL.
- All data shall be stored in normalized tables to reduce redundancy and improve consistency.
- The database shall maintain relationships between entities such as books, users, and transactions.

Data Entities

The system shall include the following primary entities:

- Books (Book ID, Title, Author, ISBN, Category, Availability)
- Users (User ID, Name, Role, Contact Details)
- Transactions (Transaction ID, Issue Date, Return Date, Fine Amount)

Data Integrity

- The system shall enforce referential integrity constraints.
- Primary and foreign keys shall be used to maintain relationships.

7.2 Legal and Regulatory Requirements

The LMS must comply with applicable legal and institutional policies regarding data usage and privacy.

- User data shall be handled in accordance with data protection regulations.
- The system shall ensure that personal information is not disclosed to unauthorized parties.
- Copyright rules shall be respected when managing digital resources.

7.3 Internationalization and Localization Requirements

Although the initial version of the system is designed for a specific region, it should support future expansion.

- The system shall support multiple languages in future versions.
- Date and time formats shall be adaptable based on regional settings.
- Currency formats (for fines) shall be configurable.

7.4 Scalability and Future Enhancements

The system shall be designed to accommodate future growth and feature expansion.

- The system architecture shall support the addition of new modules.
- Future enhancements may include mobile application support.
- Integration with external systems (e.g., digital libraries) may be implemented.

7.5 Data Backup and Recovery

- The system shall perform periodic backups of all critical data.
- Backup data shall be stored securely and be recoverable in case of system failure.
- Recovery procedures shall ensure minimal downtime.

7.6 Reusability Requirements

- The system shall be developed using modular components to allow reuse in other projects.
- Code components such as authentication modules may be reused in future systems.

Appendix A: Glossary

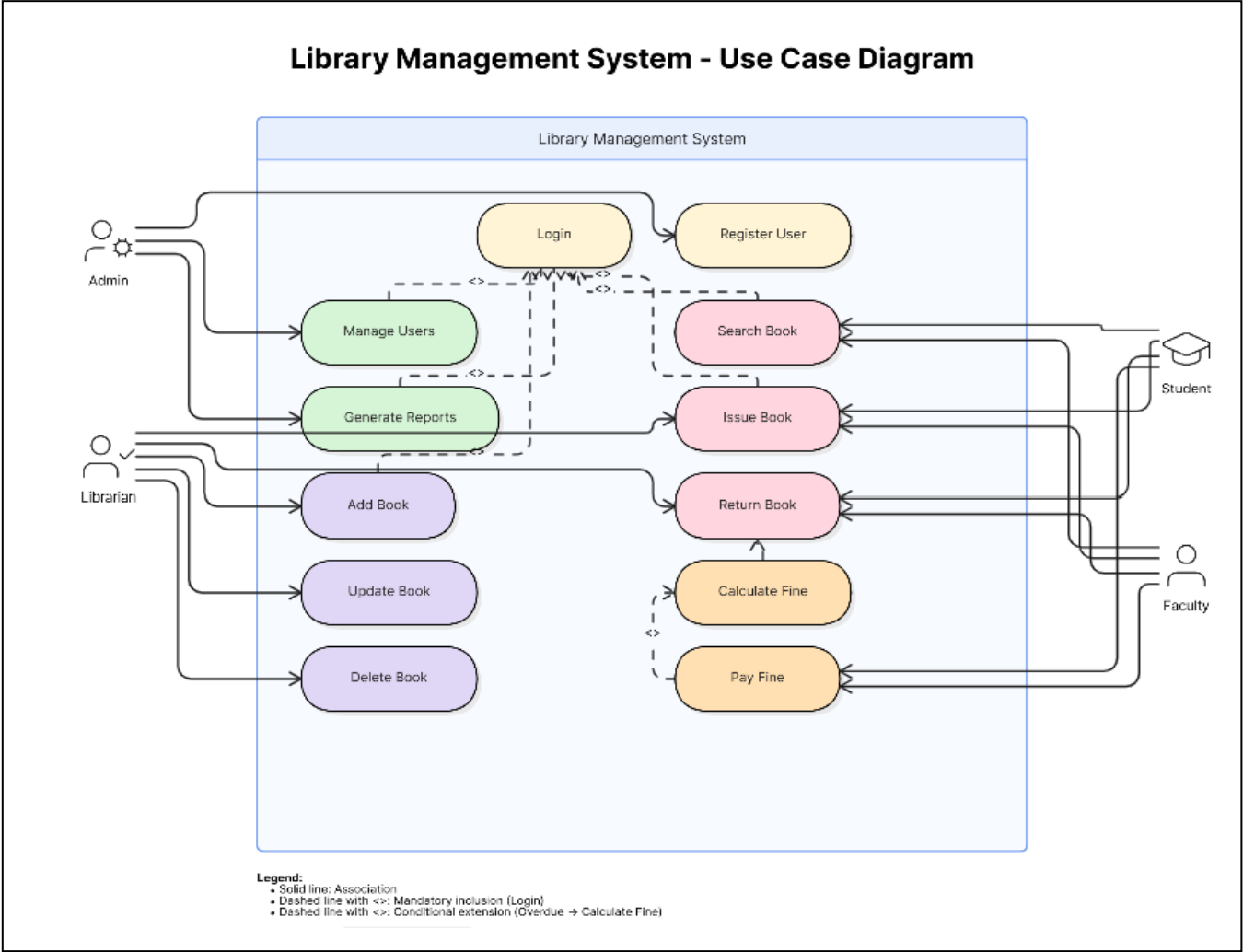
This appendix defines important terms, abbreviations, and acronyms used throughout this document.

Term	Definition
LMS	Library Management System
Admin	System administrator with full access
Librarian	User responsible for managing library operations
User	General system user (student or faculty)
ISBN	International Standard Book Number
RDBMS	Relational Database Management System
GUI	Graphical User Interface
Authentication	Process of verifying user identity
Authorization	Granting access rights to users
Transaction	Process of issuing or returning a book
Fine	Penalty charged for late return of books

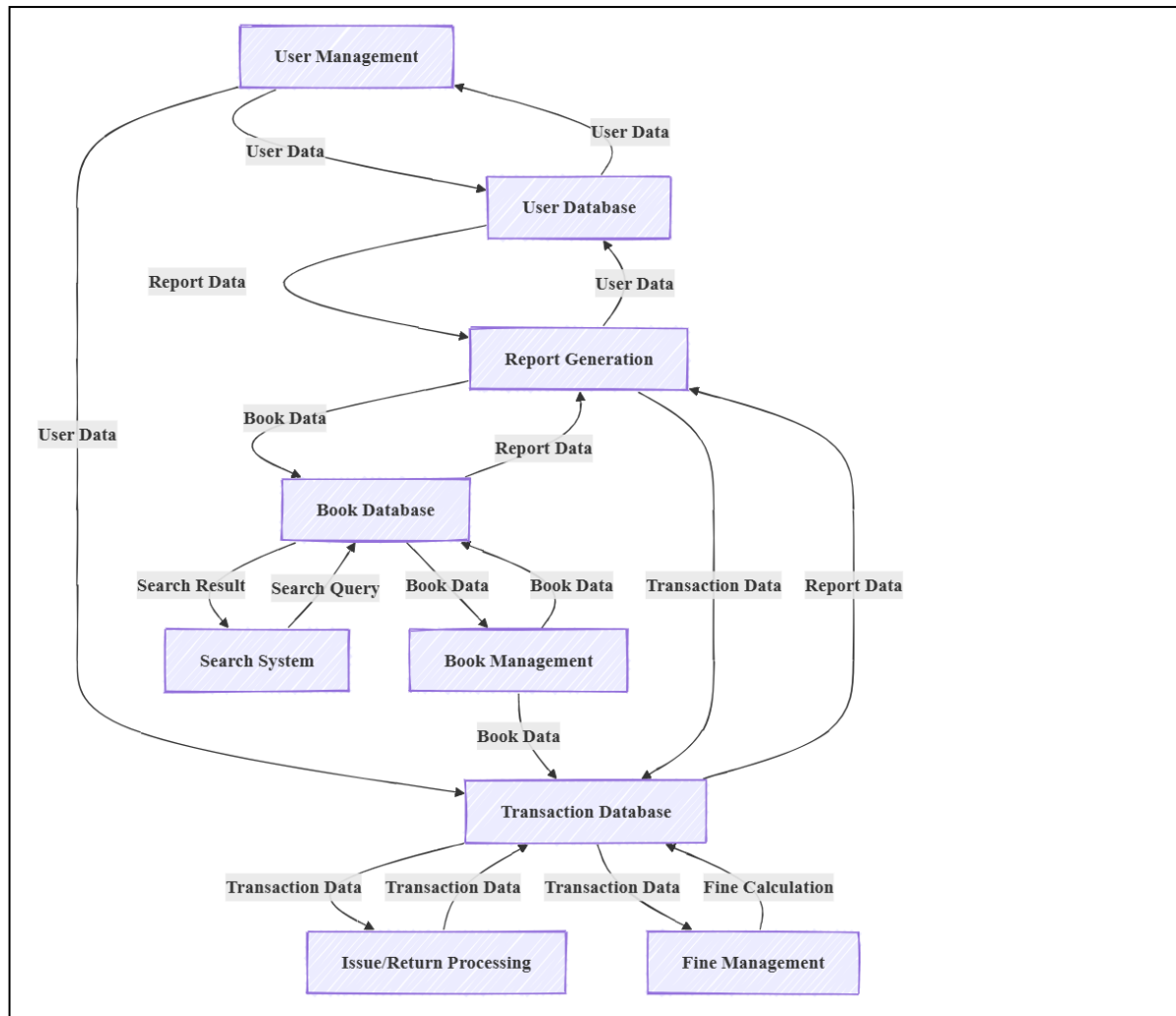
Appendix B: Analysis Models

This appendix includes diagrams and models used to represent the system’s structure and behavior.

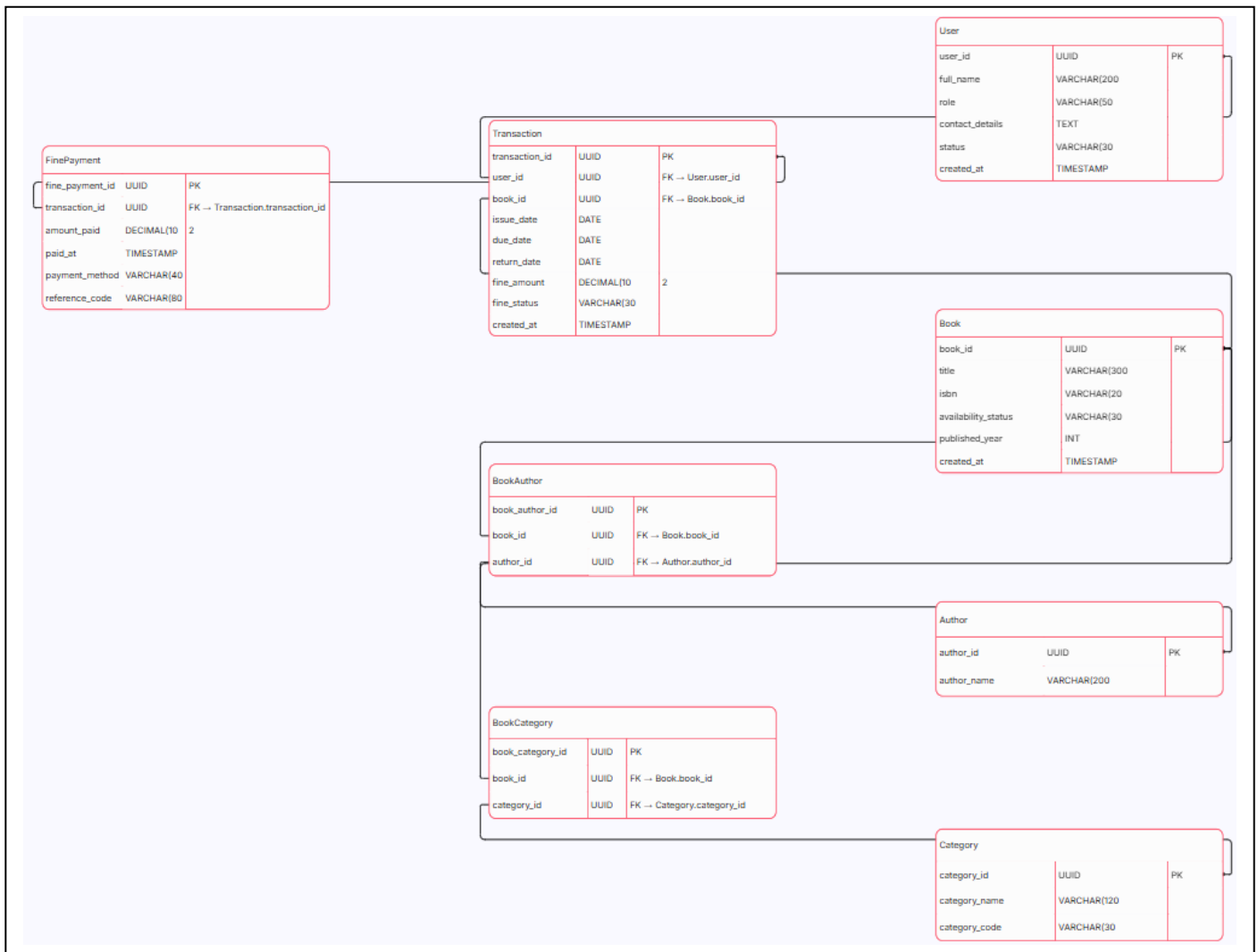
B.1 Use Case Diagram



B.2 Data Flow Diagram (DFD)



B.3 Entity-Relationship (ER) Diagram



Appendix C: To Be Determined List

This section lists items that are not yet finalized and require further clarification.

TBD ID	Description	Status
TBD-01	Final fine amount per day	Pending
TBD-02	Maximum borrowing limit	Pending
TBD-03	Notification system (SMS/Email)	Under consideration
TBD-04	Integration with external systems	TBD
TBD-05	Multi-language support details	TBD